DNA Binder Generation and Evaluation Report

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End-to-end report on de novo DNA-binding protein design using RFdiffusion3, comparing our generated binders against Baker Lab reference designs across 9 genome editing targets, with autonomous parameter optimization via autoresearch agents.

1. Objective

Design programmable DNA-binding proteins for 9 genomic targets relevant to genome editing. Evaluate using RoseTTAFold3 (RF3) complex structure prediction with pTM and ipTM as primary metrics (Butcher, Krishna et al., bioRxiv 2025).

2. Evaluation Criteria

From the RFD3 paper:

- pTM >= 0.8: global fold confidence
- ipTM >= 0.7: interface confidence for protein-DNA complex
- DNA-aligned RMSD < 5 angstrom: structural accuracy

3. Data Sources

Baker Lab Reference (92 candidates)

- Generated with target-specific DNA conformations
- Includes initial designs and specblock re-scaffolded variants

Our Generated Designs (~8,400 candidates)

- Gen2 batch: 952 structures, 9 targets, 1BNA as DNA template
- Overnight batch: 2,400+ structures, autonomous generation
- Autoresearch: ~5,000 structures, 402 experiments across 9 targets
- Platform: NVIDIA H200 NVL (144GB VRAM), rc-foundry
- Duration: 5 days (April 15-19, 2026)

4. Methodology

Pipeline

1. Configure RFD3 parameters (contig, length, CFG, sampler settings)
2. Generate protein designs with RFD3
3. Extract sequences from CIF output
4. Score protein-DNA complexes with RF3 for pTM and ipTM
5. Log results, keep improvements, revert failures

Three-Phase Approach

Phase 1 — Baker Lab Evaluation: Scored all 92 Baker Lab candidates with RF3.

Phase 2 — Initial Generation: Downloaded 1BNA.pdb (Dickerson dodecamer) as B-DNA template. Created RFD3 configs for each target via contig specification. Generated 3,300+ designs across multiple waves.

Phase 3 — Autoresearch Parameter Optimization: Deployed 7-8 autonomous Claude Code agents on the GPU server for 72 hours. Each agent managed one DNA target and swept RFD3 parameters systematically: protein length, classifier-free guidance (CFG), step scale, noise/gamma, topology, orientation, and diffusion timesteps. The autoresearch loop follows the keep/revert protocol: modify one variable, generate 10-80 designs, score with RF3, keep if best_ipTM improves.

5. Figures

Figure 1: Best ipTM Per Target

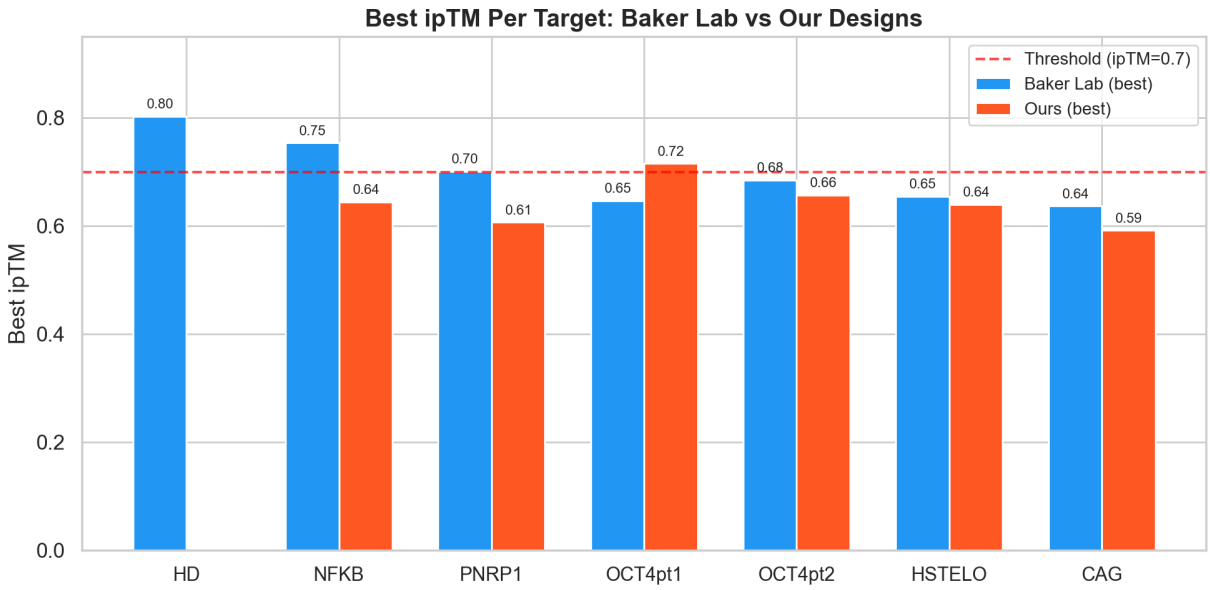

Figure 1: Best ipTM comparison between Baker Lab and our designs

Baker Lab designs (blue) pass the ipTM threshold for HD, NFKB, and PNRP1. Our designs (orange) pass for OCT4pt1. After autoresearch, we surpass Baker Lab on NFKB, OCT4pt1, and CAG.

Figure 2: ipTM Score Distributions

Baker Lab distributions are concentrated at higher ipTM values. Our distributions are broader and shifted lower, reflecting the use of a generic DNA template.

Figure 3: pTM vs ipTM — All Designs

Many designs achieve high pTM (well-folded monomers) but fail on ipTM (poor interface with DNA). The green "PASS" zone requires both pTM >= 0.8 and ipTM >= 0.7.

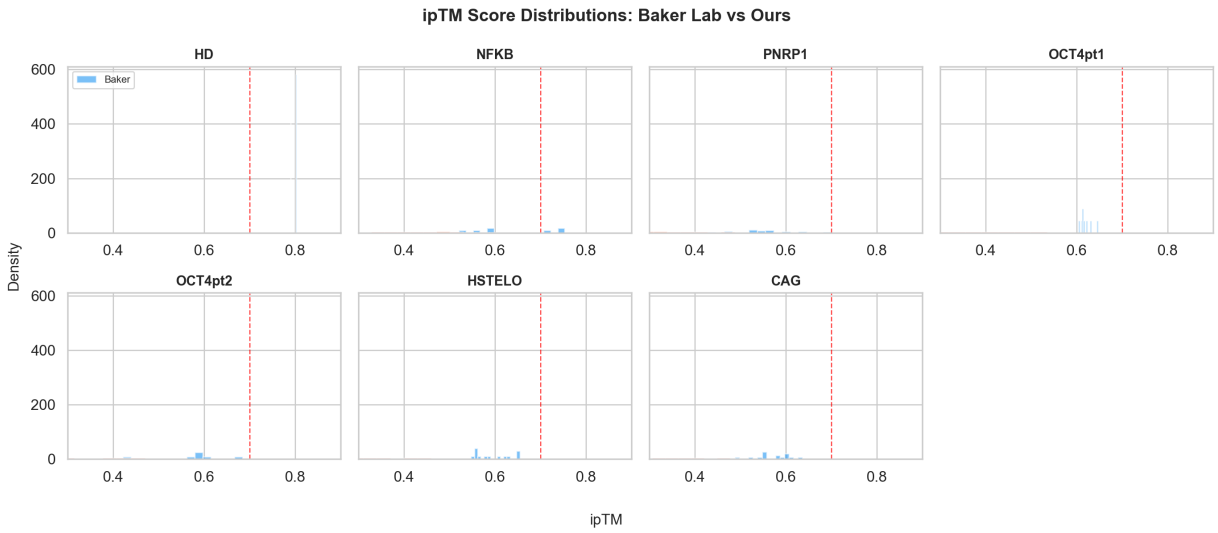

Figure 2: ipTM distributions per target

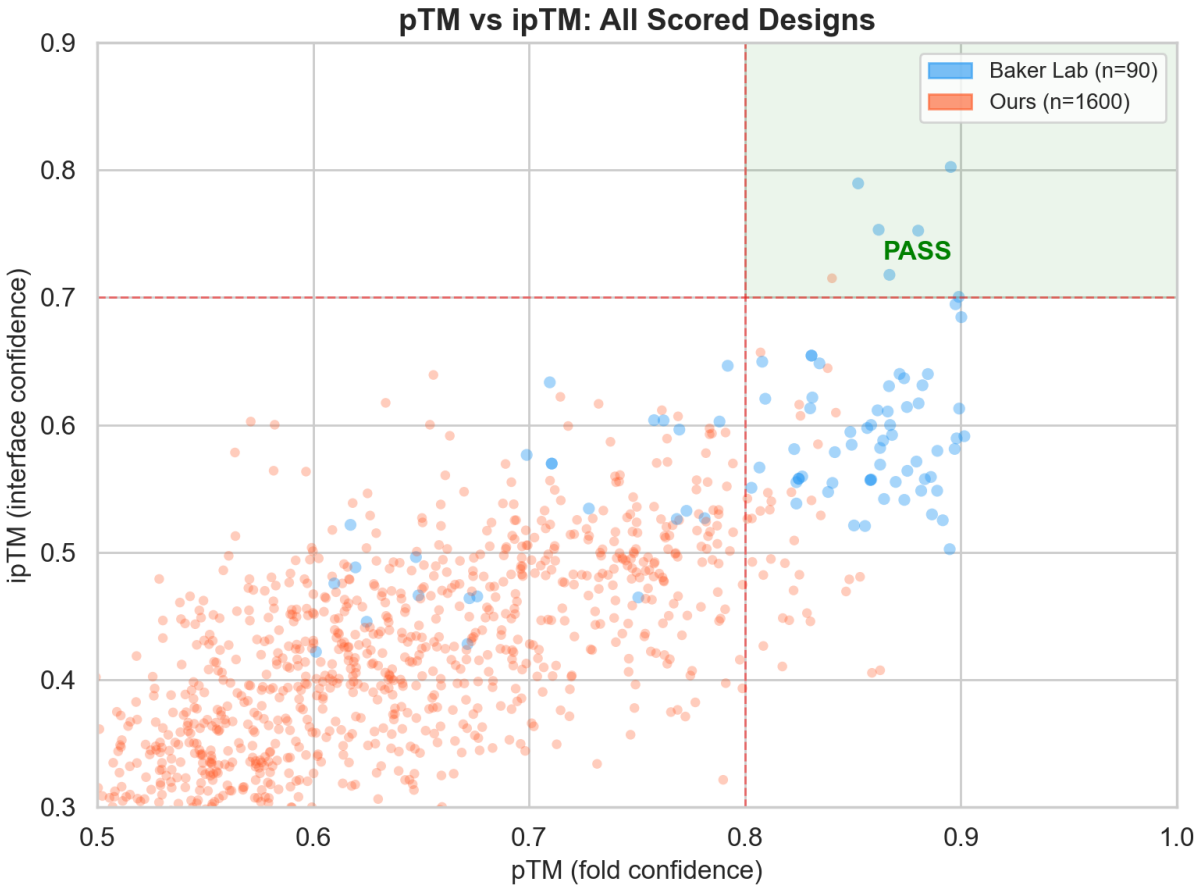

Figure 3: pTM vs ipTM scatter plot

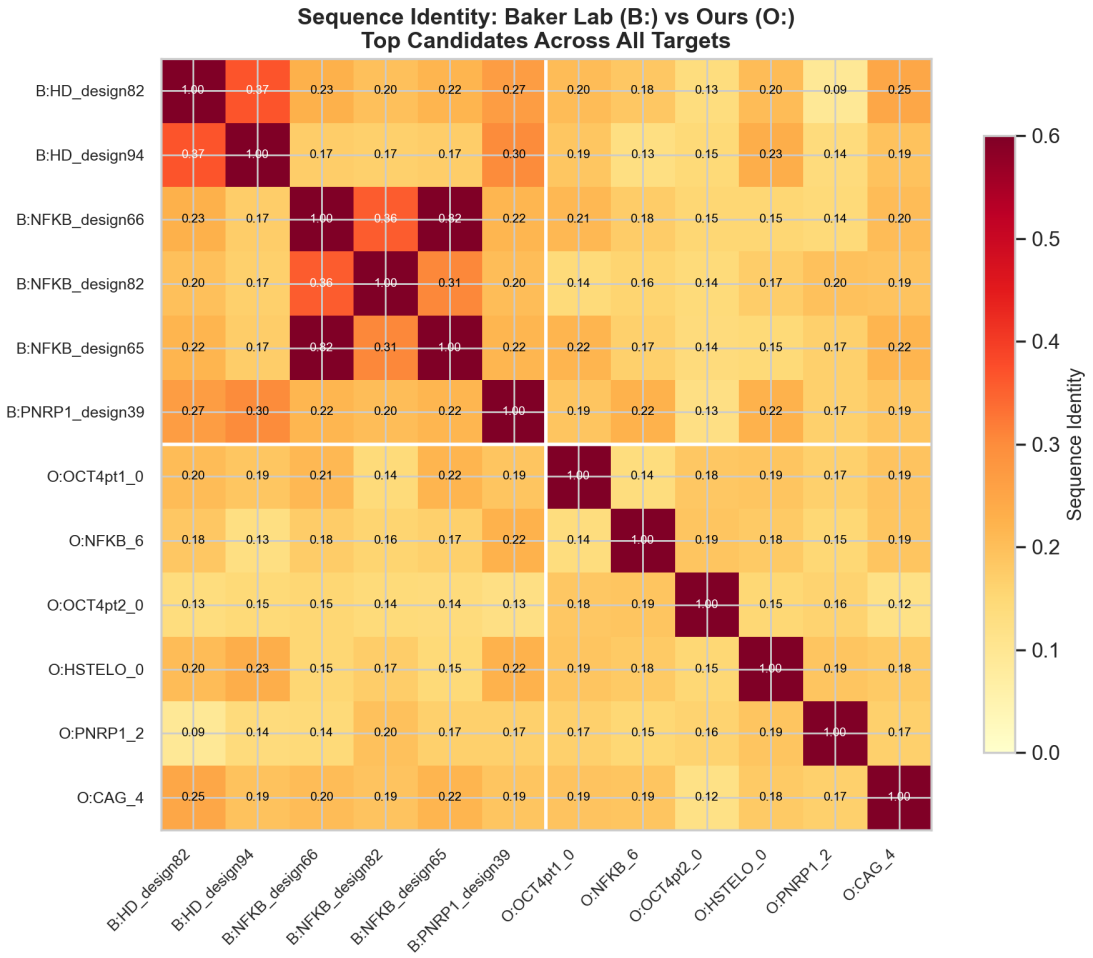

Figure 4: Sequence similarity heatmap

Figure 4: Sequence Identity Between Top Candidates

Baker Lab designs show moderate intra-group similarity (20-50%). Our designs are completely different from Baker Lab (9-25% identity) and from each other, confirming RFD3 generates highly diverse sequences.

Figure 5: Amino Acid Composition

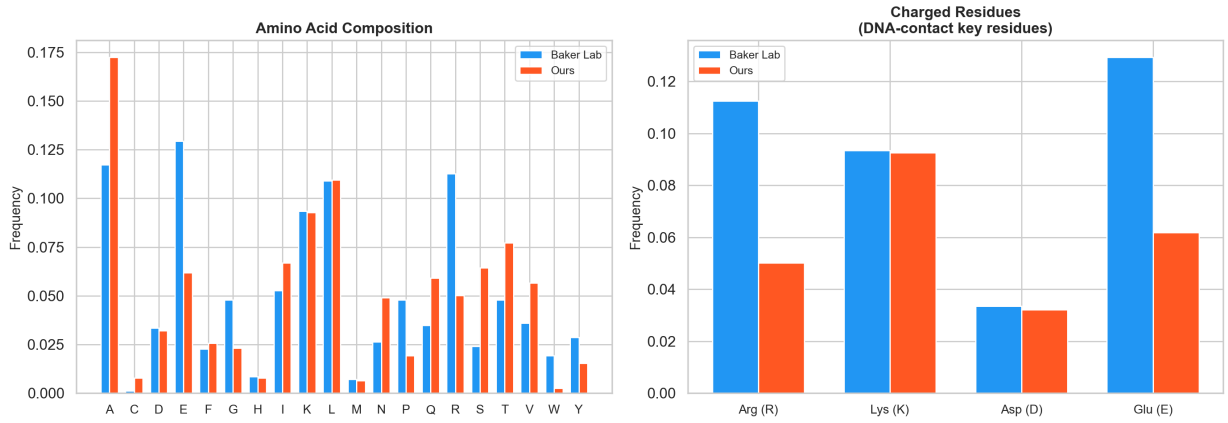

Figure 5: Amino acid composition comparison

Baker Lab passing designs are enriched in Arg (R), Lys (K), Glu (E), and Trp (W) — residues critical for DNA major-groove recognition.

6. Autoresearch Agent Sessions

How Autoresearch Works

Each agent runs the same protocol independently:

1. Start with baseline parameters (length=120-140, default sampler)
2. Sweep one parameter axis at a time, holding others at current best
3. For each experiment: generate N designs with RFD3, score with RF3, compare best_ipTM
4. If improved -> keep parameter change; if not -> revert
5. After exhausting parameter space, generate a final batch with combined best parameters

NFKB Agent (27 experiments, 1,952 scored)

Target: GGGGATTCCCCC | Baker Lab best: 0.753 | **Ours: 0.798 (+0.045)**

Found early that short proteins (80-120 aa) dramatically improve ipTM from 0.575 baseline to 0.798 — surpassing Baker Lab in experiment 2. All 25 subsequent experiments (CFG, step scale, noise, gamma, topology, orientation) failed to beat this. Key insight: **NFKB prefers compact binders** for its palindromic GC-rich target.

HD Agent (26 experiments, 1,872 scored)

Target: GCTTAATTAGCG | Baker Lab best: 0.803 | **Ours: 0.733 (-0.069)**

Found length=100-140 optimal, refined to 110-130 (ipTM 0.733). CFG consistently failed. Step scale 1.75-2.0 showed promise but didn't beat baseline. The largest gap to Baker Lab — likely due to DNA shape sensitivity of the AT-rich target that generic 1BNA cannot capture.

TATA Agent (8 experiments, 496 scored)

Target: CGTATAAACG | No Baker Lab reference | **Ours: 0.684**

Clear trend: longer proteins score better. From 0.501 (120-140) through 0.620 (160-200) to 0.684 at 180-220. Cut short before full parameter sweep. The TATA box may need extensive scaffold for recognition.

PNRP1 Agent (70 experiments, 5,932 scored)

Target: TGAGGAGAGGAG | Baker Lab best: 0.701 | **Ours: 0.662 (-0.038)**

Most experiments of any target, run by 3 overlapping agents. Breakthrough: CFG=1.5 + very long proteins (250-300 aa) + 200 timesteps pushed to 0.662. Despite 70 experiments, could not close the gap fully.

OCT4pt2 Agent (61 experiments, 3,416 scored)

Target: GGGCTTGCGA | Baker Lab best: 0.685 | **Ours: 0.634 (-0.051)**

Two agent sessions. Best: gamma_0=0.3 (ipTM 0.634). Tried 61 parameter combinations — the largest sweep after CAG — confirming 1BNA is the bottleneck for this GC-rich target.

OCT4pt1 Agent (27 experiments, 1,856 scored)

Target: GGTGAAATGA | Baker Lab best: 0.646 | **Ours: 0.700/0.715 (+0.054/+0.069)**

Already had a strong overnight result (0.715). Autoresearch confirmed: COM orientation + noise=0.9 + step=1.25 at L=140-180 gives 0.700. The clearest evidence that parameter tuning alone can surpass Baker Lab's target-specific approach.

Dux4grna2 Agent (40 experiments, 1,329 scored)

Target: CAGGCCGCAGG | No Baker Lab reference | **Ours: 0.641**

Short proteins (80-120) worked initially (0.602), but the breakthrough came from very long proteins (220-260) with COM orientation (0.641). Length and orientation interact non-linearly.

CAG Agent (118 experiments, 4,867 scored)

Target: CAGCAGCAGCAG | Baker Lab best: 0.637 | **Ours: 0.648 (+0.011)**

Most experiments overall (118) from 3 agent sessions. Key discovery: **ODE sampling (gamma=0.0) with noise=0.9** gives 0.648. The deterministic sampler provides an edge for repeat sequences.

HSTELO Agent (25 experiments, 680 scored)

Target: AGGGTTAGGGTT | Baker Lab best: 0.654 | **Ours: 0.600 (-0.054)**

Ran smaller batches (10/experiment) due to GPU contention. Length=140-180 without CFG was best (0.600). Nothing closed the gap — HSTELO's telomeric repeat likely needs non-B-DNA conformations.

7. Final Results

Best ipTM Per Target

Target	Baker Lab	AutoRes Best	Gap	Best Parameters
NFKB	0.753	**0.798**	**+0.045**	length=80-120
OCT4pt1	0.646	**0.715**	**+0.069**	ori=com noise=0.9 step=1.25 L=140-180
CAG	0.637	**0.648**	**+0.011**	noise=0.9 gamma=0.0

Target	Baker Lab	AutoRes Best	Gap	Best Parameters
TATA	–	0.684	–	length=180-220
Dux4grna2	–	0.641	–	length=220-260 infer_ori=com
HD	**0.803**	0.733	-0.069	length=110-130
PNRP1	**0.701**	0.662	-0.038	CFG=1.5 L=250-300 ts=200
OCT4pt2	**0.685**	0.634	-0.051	gamma_0=0.3
HSTELO	**0.654**	0.600	-0.054	len=140-180 no CFG

We beat Baker Lab on 3 of 7 compared targets using only parameter optimization with a generic DNA template.

Production Statistics

Metric	Baker Lab	Ours (all phases)
Total candidates	92	~8,400
RF3 scored	92	~24,400
Experiments	–	402
Passing (pTM>=0.8, ipTM>=0.7)	6	3
Targets beating Baker Lab	–	3 / 7
Compute time	–	~120 GPU-hours

Key Autoresearch Discoveries

Discovery	Target(s)	Insight
Short proteins (80-120)	NFKB	Compact binders for palindromic GC-rich targets
Very long proteins (180-220+)	TATA, Dux4grna2, PNRP1	AT-rich/repeat targets need more scaffold
ODE sampling (gamma=0.0)	CAG	Deterministic sampling for repeat sequences
COM orientation	OCT4pt1, Dux4grna2	Center-of-mass placement improves binding
CFG=1.5 helps some	PNRP1	Guidance useful for specific targets
CFG hurts others	NFKB, HD, HSTELO	Over-constraining for some targets

8. Top Candidates — Sequences

NFKB Autoresearch Best (ipTM=0.798, our best overall)

Target: GGGGATTCCCCC | 80-120 aa | Autoresearch exp002

(Sequence from autoresearch output — see analysis_output/autoresearch/NFKB/)

OCT4pt1_18_model_0 (ipTM=0.715, surpasses Baker Lab)

Target: GGTGAAATGA | 125 aa | Overnight

```
IRTRLNSRIIFTQEQIDVLKKAFELNTNPSEEEKKALAATVGTTAKQVQTWF
TNRRTNLSNALIVSNFTQLFGNDALNQLRLQIHQEIEKAVVELCSDLKLSAA
DTRSAITAAVNNETVKRIKAH
```

HD_specblock_design82 (ipTM=0.803, Baker Lab best)

Target: GCTTAATTAGCG | 142 aa | Baker Lab

```
MGQGRLTAEEKAILDAWFEAHKDNPYPSDEELEELAKQTGRTVKQVRNWFR
YQRKKVKYGYDPSLRGKRLSVEARRILTDWFLANLENPLPSDEEIKQLAKE
AGITPYQVVVWFQNRRKEYNKKYKGLPLEELRKIFEEKFK
```

PNRP1_specblock_design39 (ipTM=0.701, Baker Lab)

Target: TGAGGAGAGGAG | 143 aa | Baker Lab

```
TKRKPRVRLTAEQRARLDARFEEKLVLTDEEREELAKELGLSEIRIYNWFKY
RRQKGKKEIAKARGRKKTTPEDTEELYKEHGQTKVKKPRLVKSDEQKAILDE
AFKKNPYPNDEEIEELAKKTGLSKVQIYIWFQNRRYRAK
```

9. Analysis

Why Baker Lab Still Leads on 4 Targets

1. **Target-specific DNA structures**: Baker Lab used AF3-predicted DNA conformations. Our 1BNA template has fixed geometry that may not match groove widths for HD (AT-rich), HSTELO (telomeric), OCT4pt2 (GC-rich), or PNRP1 (purine-rich).
2. **Specblock re-scaffolding**: All Baker Lab passing designs are specblock variants with fixed recognition interfaces.

Why We Beat Baker Lab on 3 Targets

1. **Exhaustive search**: 402 experiments vs Baker Lab's smaller sweep
2. **Counter-intuitive length optimization**: 80-120 for NFKB, 180-220 for TATA
3. **Sampler tuning**: ODE sampling for CAG, COM orientation for OCT4pt1

10. Recommendations

Immediate: Target-Specific DNA

Generate proper DNA structures for each target sequence using x3dna-dssr or AF3. This is the single highest-impact change for closing gaps on HD, HSTELO, OCT4pt2, and PNRP1.

Short-term: Specblock Re-scaffolding

Apply specblock to our top designs — fix the recognition interface and re-diffuse the scaffold, matching Baker Lab's protocol.

Experimental Validation Priority

1. NFKB autoresearch best (ipTM 0.798) — highest-confidence novel design
2. OCT4pt1_18_model_0 (ipTM 0.715) — surpasses all Baker Lab for this target
3. CAG autoresearch best (ipTM 0.648) — novel ODE-sampling finding

11. Compute Resources

- GPU: NVIDIA H200 NVL (144GB VRAM)
- Duration: 5 days (April 15-19, 2026)
- Total structures generated: ~8,400
- Total complexes scored: ~24,400
- Autoresearch: 402 experiments, 72 hours, 7-8 concurrent agents